Fowler (2024) on the Japanese commuting zones

August 27, 2026

What this deck reports

- Every table and figure of Fowler (2024) that the Japanese data supports, in the order that paper presents them.
- Delineations for 1980 to 2020; this deck shows the 2010 against 2020 comparison, the same decade spacing as the source paper.
- The wage-correlation block, that paper's Figure 2 and the Connection rows of its Table 2, waits on the Basic Survey on Wage Structure microdata.
- Buttons in the lower right lead to the appendix and back.

Source paper: `https://doi.org/10.1038/s41597-024-03829-5`

The delineation

For municipalities i and j with commuting flows f_{ij} and resident labour forces W_i, the proportional flow is

$$p_{ij} = \frac{f_{ij} + f_{ji}}{\min(W_i, W_j)}, \qquad p_{ij} \leq 0.999,$$

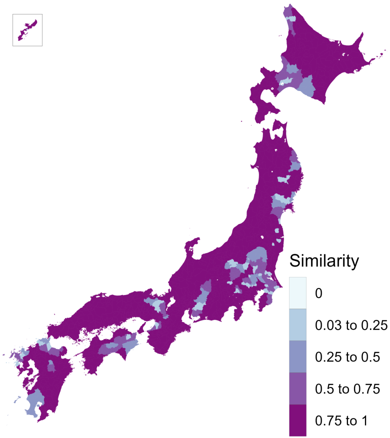
and the dissimilarity is $1 - p_{ij}$ with a zero diagonal.

- Average-linkage hierarchical clustering, cut at a fixed tree height of 0.977, with merges restricted to clusters that adjoin on land.
- Commuting matrix: residents mainly working, ordinary households, the only definition available for 2020.
- 1,656 municipalities: the four main islands and the Okinawa main island, plus every municipality joined to one by a permanent road link.
- The twenty-three special wards of Tokyo count as one, as the ordinance-designated cities do. Split, a suburb's commute reads as twenty-three weak links.
- Cores and urban areas for the fit statistics: the Urban Employment Area central cities and their areas.

▸ The offshore islands ▸ Departure from the pipeline

Similarity between the 2010 and the 2020 delineations

Figure 1 of Fowler (2024)



Jaccard similarity: the overlap of a municipality's two commuting zones over their union. Mean 0.86.

▸ Every comparison pair ▸ Zones held at 1980

Diagnostic statistics

Table 1 of Fowler (2024)

	2010	2020
Municipalities in the delineation	1,656	1,655
Commuting zones	231	223
Single-municipality zones	9	10
Municipalities in the largest zone	27	29
Non-contiguous zones	0	0
Smallest zone, resident labour force	530	475
Average zone, resident labour force	202,694	207,532
Largest zone, resident labour force	4,929,463	5,624,897
Smallest zone area (sq. km)	184	100
Average zone area (sq. km)	1,574	1,630
Largest zone area (sq. km)	5,744	5,744
Compactness of zones	0.30	0.30

Compactness is the Polsby–Popper ratio, $4\pi A/P^2$ of the dissolved zone, averaged across zones.

▸ Counts, all nine years ▸ Size and shape, all nine years

Fit statistics

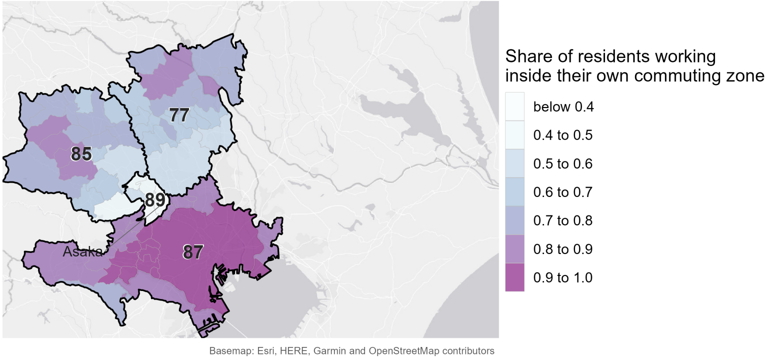
Table 2 of Fowler (2024)

	2010	2020
Core		
Urban areas split across zones	31	33
Zones containing a core	72.7%	67.7%
Residents working in a core of their zone	52.3%	51.9%
Workforce living in a core of their zone	47.9%	47.7%
Connection		
Minimum wage correlation	pending	
Mean wage correlation	pending	
Mean wage correlation, multi-municipality zones	pending	
Containment		
Minimum contained	44.5%	40.5%
Mean contained	90.0%	90.3%
Share of the labour force contained	85.8%	86.2%

▸ Containment, all nine years ▸ Core measures, all nine years

The commuting zone with the lowest mean containment

Figure 3 of Fowler (2024)



Six municipalities in southern Saitama at 0.41, between the Tokyo zone and the zones beyond.

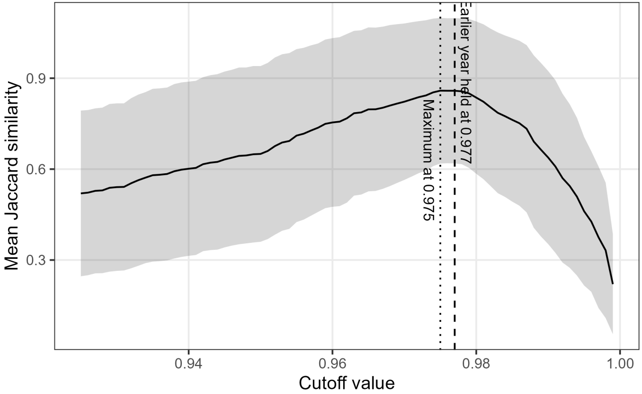
▸ The same map at 0.980

Technical validation

- The cutoff. It is inherited rather than derived. Does the similarity between delineations pick out a better value?
- The contiguity constraint. What imposing it costs, and what the delineation looks like without it.
- Split urban areas. Externally defined urban areas cut across more than one commuting zone.

Similarity as a function of the cutoff

Figure 4 of Fowler (2024)



Holding 2010 at 0.977, the peak is 0.86 at 0.975, and the curve stays within 0.005 of it from 0.975 to 0.978, spanning 232 down to 218 zones.

▸ The two cutoffs compared ▸ The control pairs

The delineation with and without the contiguity constraint

2020, at the 0.977 cutoff

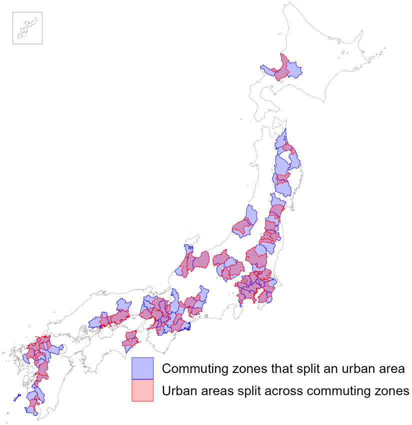
	Without the constraint	With the constraint
Commuting zones	223	223
Zones holding a detached municipality	1	0
Minimum contained	40.5%	40.5%
Mean contained	90.4%	90.3%
Share of the labour force contained	86.1%	86.2%

The constraint moves 66 of 1,655 municipalities, and the two delineations agree at 0.99.

▸ Without the constraint ▸ All nine census years

Urban areas split across commuting zones

Figure 6 of Fowler (2024)



33 of the 208 Urban Employment Areas in 2020, concentrated along the Pacific corridor.

Core measures, all nine years

Where everything lives

On the repo: `https://github.com/daisukeadachi/commuting_zone_japan/`

- `validation/notes/replication_results.md` — the written results, with every number in this deck in context.
- `validation/notes/island_coverage.md` — the offshore islands, and what covering them changes.
- `validation/notes/zone_crosswalk.md` — the published crosswalk, column by column.
- `validation/notes/method_crosscheck.md` — the delineation formula across four implementations.
- `validation/notes/core_definition.md` — why the Urban Employment Area supplies the cores.
- `validation/output` — the tables behind every slide, as comma-separated files.

Appendix

- The offshore islands, and the departure from the existing pipeline.
- Every comparison pair, and what holding the zones at 1980 costs.
- Diagnostic statistics, zone size and shape, 1980 to 2020.
- Containment and the Core measures, 1980 to 2020.
- The lowest-containment map at the other cutoff.
- The two cutoff anchors compared, and the control-pair curves.
- The delineation without the contiguity constraint.

The offshore islands

Year	Island municipalities	Island zones	Resident labour force	Share of the labour force	Mean contained
1980	63	56	293,232	0.63%	98.4%
1985	63	55	287,304	0.60%	97.7%
1990	63	53	275,065	0.53%	98.3%
1995	63	53	272,929	0.51%	98.4%
2000	62	52	263,150	0.49%	98.5%
2005	63	52	250,311	0.49%	98.4%
2010	63	52	233,419	0.50%	98.5%
2015	63	52	227,039	0.49%	98.4%
2020	63	52	215,526	0.46%	98.5%

Six of the 52 zones in 2020 hold more than one municipality: Rishiri, Shodoshima, Tanegashima, Amami Oshima, Tokunoshima and Okinoerabu.

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One departure from the existing pipeline

The existing Japanese pipeline joins the flow file to its own transpose in a way that keeps a pair only when the flow from i to j is recorded, so a pair connected only in the reverse direction enters as unconnected.

- The matrix it clusters is asymmetric; this replication uses the symmetric numerator the source method specifies.
- Between 1.2 and 4.4 percent of pairs are affected in each year, moving up to four zones and up to 108 municipalities.
- Measured year by year in `validation/notes/method_crosscheck.md`.

Every comparison pair

Pair	Role	Municipalities	Mean	Labour-force weighted
1980 against 1990	appendix	1,656	0.78	0.78
1990 against 2000	appendix	1,656	0.84	0.87
2000 against 2010	appendix	1,656	0.86	0.86
2005 against 2015	baseline	1,651	0.85	0.85
2010 against 2015	baseline	1,651	0.89	0.88
2010 against 2020	baseline	1,655	0.86	0.83

The 2010 to 2020 comparison sits below both pre-pandemic controls but above the decade pairs of the 1980s and 1990s.

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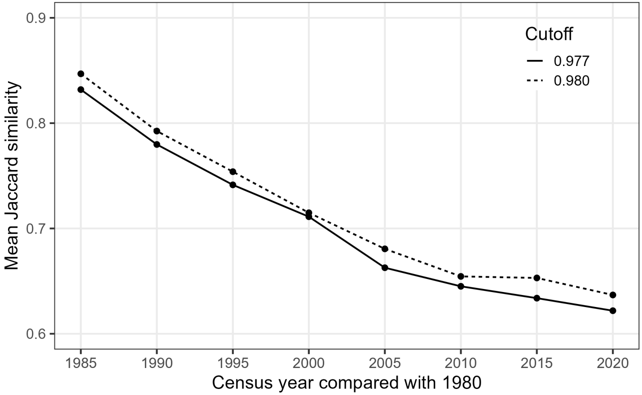
What holding the zones at 1980 costs



Both years cut at the same height. The crosswalk carries a column for every anchor year.

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Diagnostic statistics, 1980 to 2020

Year	Municipalities	Commuting zones	Single-municipality	Largest zone	Non-contiguous
1980	1,656	385	72	31	0
1985	1,656	343	56	29	0
1990	1,656	307	35	29	0
1995	1,656	282	26	28	0
2000	1,656	261	16	30	0
2005	1,656	240	12	27	0
2010	1,656	231	9	27	0
2015	1,651	223	9	29	0
2020	1,655	223	10	29	0

Counts at the 0.977 cutoff. Non-contiguous zones are those holding a municipality with no neighbour in its own zone.

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Zone size and shape, 1980 to 2020

Year	Resident labour force			Area (sq. km)			Compactness
	Smallest	Average	Largest	Smallest	Average	Largest	
1980	286	120,006	4,761,443	16	944	3,464	0.33
1985	650	139,881	5,482,720	24	1,060	3,839	0.33
1990	788	168,711	5,927,239	24	1,184	4,429	0.32
1995	723	189,076	5,300,358	24	1,289	4,429	0.32
2000	523	202,706	6,000,065	109	1,393	4,429	0.31
2005	295	210,644	5,439,539	100	1,515	4,733	0.31
2010	530	202,694	4,929,463	184	1,574	5,744	0.30
2015	479	207,690	5,031,627	184	1,627	5,744	0.30
2020	475	207,532	5,624,897	100	1,630	5,744	0.30

Population is the resident labour force carried in the commuting matrix. Compactness is the Polsby–Popper ratio.

Containment, 1980 to 2020

Year	Minimum	Mean	Labour-force weighted mean	Share of the labour force	Mean work contained
1980	42.9%	91.0%	88.8%	88.7%	94.5%
1985	23.0%	90.3%	88.5%	88.2%	93.9%
1990	21.2%	89.9%	87.8%	87.3%	93.4%
1995	20.9%	89.1%	86.7%	86.3%	92.6%
2000	40.6%	89.9%	87.0%	86.6%	92.3%
2005	41.4%	89.8%	86.8%	86.1%	92.2%
2010	44.5%	90.0%	86.5%	85.8%	92.1%
2015	49.2%	90.3%	86.4%	85.6%	91.7%
2020	40.5%	90.3%	87.0%	86.2%	91.4%

The minimum and the mean are taken over commuting zones; the share of the labour force is a single national ratio.

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Core measures, 1980 to 2020

Year	Urban Employment Area				District population 10,000 or more		
	With core	Work in	Live in	Split	With core	Work in	Live in
1980	55.6%	57.9%	53.5%	50	60.5%	73.9%	72.4%
1985	–	–	–	–	64.1%	74.1%	72.8%
1990	61.6%	55.9%	51.2%	46	68.4%	74.6%	73.4%
1995	65.2%	54.5%	49.9%	50	72.0%	74.4%	73.4%
2000	68.6%	54.3%	49.8%	55	74.7%	74.8%	73.9%
2005	70.8%	52.8%	48.5%	39	77.1%	74.4%	73.6%
2010	72.7%	52.3%	47.9%	31	77.9%	74.0%	73.2%
2015	74.0%	51.3%	47.0%	34	78.5%	73.6%	72.9%
2020	67.7%	51.9%	47.7%	33	73.5%	74.4%	73.9%

The district rule marks a municipality as a core on its densely inhabited district population alone, using no commuting data. There is no 1985 Urban Employment Area delineation.

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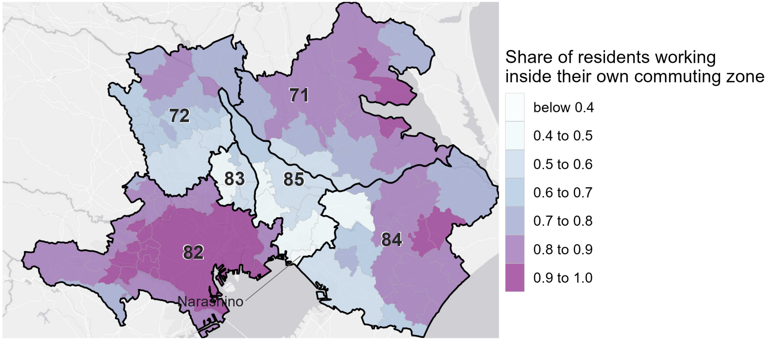
The lowest mean containment at the 0.980 cutoff



Basemap: Esri, HERE, Garmin and OpenStreetMap contributors

Same extent as the map in the body. Zone 89 is absorbed into Tokyo, which becomes zone 82; the minimum moves to Chiba, at 0.50.

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The two cutoff anchors compared

Year	Cutoff	Commuting zones	Non-contiguous	Minimum contained	Mean contained	Share of the labour force	Areas split
2010	0.977	231	0	44.5%	90.0%	85.8%	31
2020	0.977	223	0	40.5%	90.3%	86.2%	33
2010	0.980	216	0	48.7%	90.8%	86.3%	28
2020	0.980	208	0	49.9%	90.8%	86.6%	32

Both columns are computed on the constrained delineation, so the non-contiguous count is zero throughout by construction.

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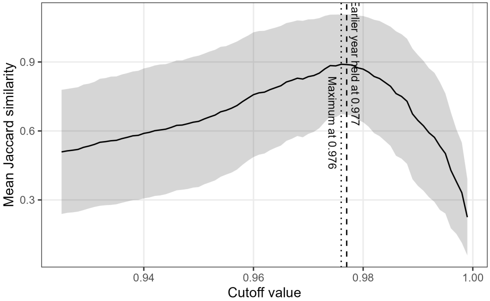
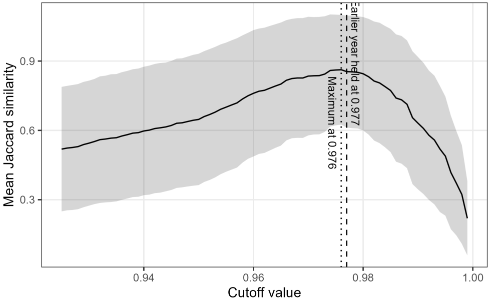
Similarity against the cutoff, the two control pairs





2010 against 2015 on the left, 2005 against 2015 on the right, both anchored at 0.977.

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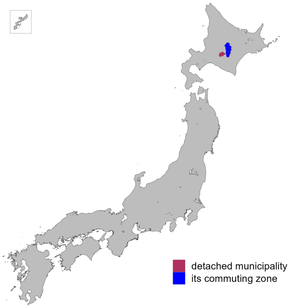
Non-contiguous municipalities without the constraint

Figure 5 and Table 3 of Fowler (2024)



Municipality	Zone	To own zone	Nearest zone	To that zone	Suggestion
Shimukappu	26	95.9%	18	2.9%	Retain

Its own zone takes 96 percent of Shimukappu's commuters and lies 991 metres away. What detaches it runs inward: 66 workers arrive from Shimizu and Shintoku.

The two delineations, 1980 to 2020

Year	Commuting zones		Detached	Changed	Mean contained	
	No constraint	Constrained	zones	municipalities	No constraint	Constrained
1980	384	385	2	36	91.1%	91.0%
1985	341	343	2	78	90.4%	90.3%
1990	307	307	1	47	89.9%	89.9%
1995	281	282	2	32	89.2%	89.1%
2000	261	261	1	4	89.9%	89.9%
2005	239	240	0	45	90.1%	89.8%
2010	231	231	0	0	90.0%	90.0%
2015	225	223	2	49	90.1%	90.3%
2020	223	223	1	66	90.4%	90.3%

Changed municipalities are those whose zone partners differ between the two delineations.

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